ARCHITECTURAL AND ENGINEERING DESIGN FOR PROPOSED LOCHAB PERIMETER WALLING

B.6.1.1 LAND SURVEY REPORT

Report by:

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17th August 2021

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SURVEYOR'S REPORT

FOR

A TOPOGRAPHIC & BOUNDARY SURVEY ON PROPERTIES
LR 7785/157-160 ALONG RUNDA DRIVE, IN RUNDA
ESTATE, NAIROBI COUNTY

FOR

THE USA EMBASSY IN KENYA
UN AVENUE ROAD
NAIROBI, KENYA

Report by: Wallace Ottaro – Bsc. Geomatic Eng (Hons), MISK (G)
Land Surveyor / Director

Date: 12th August 2021

Introduction

A report is hereby issued following a survey exercise (mentioned above) that was carried out in the month of August 2021 with instructions from the above mentioned client with the following scope;

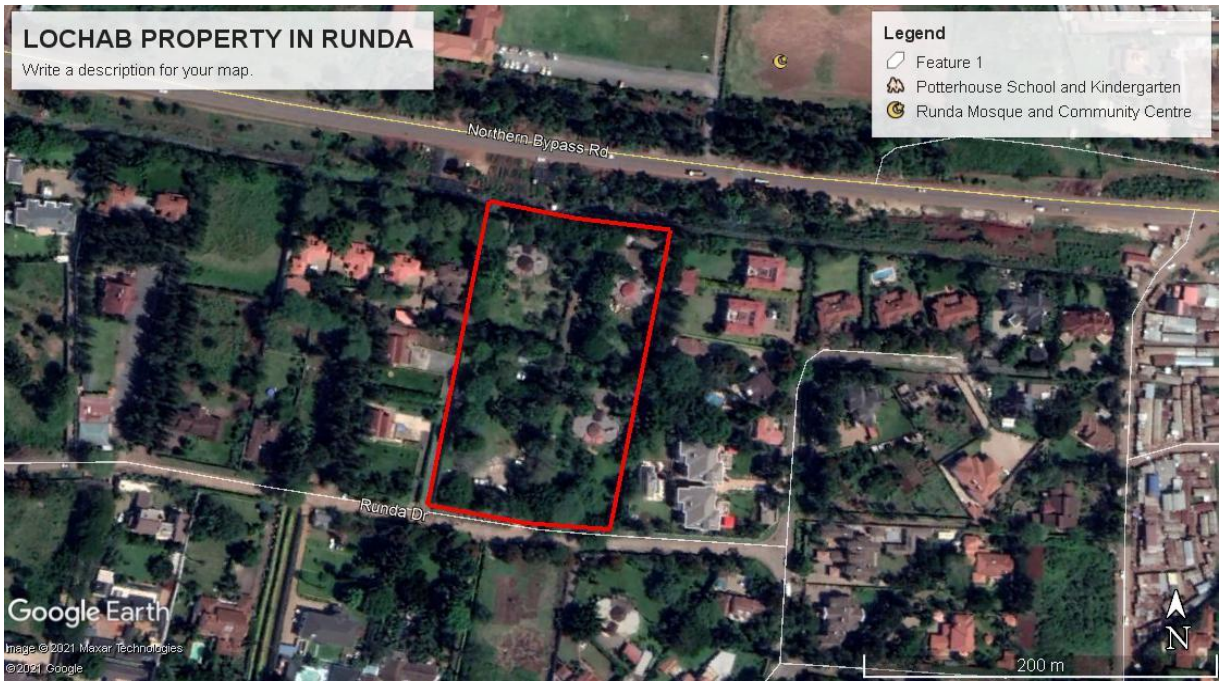
Scope of works

1. Obtaining survey reference plans of the said property from relevant authorities.

2. Conducting a boundary survey to establish the boundary beacons.

3. Conducting a Topographic Survey at the properties and processing field data to produce a contoured Topographic survey plan.

Site location



A site location image of the site courtesy of Google Earth

Methodology

1. The Boundary Survey

The survey team adopted ground survey method in executing the task while making use of the following equipment.

i. Leica Total Station
ii. KQ GEO High Precision GNSS equipment.

The LOCHAB property consists of 4 properties adjoining each other. The properties are LR 7785/ 157, 158, 159 & 160. The boundary survey was aimed at establishing the external boundary beacons of the adjoined 4 plots for purposes of constructing a perimeter wall around the 4 properties. The boundary beacons that were to be established are as follows; **[H7, H8, H9, H43, H46 & H47**]

The survey team commenced the exercise by obtaining the reference survey map of the area **FR 131/69** from the Lands registry located at the Survey of Kenya Headquarters in Ruaraka, Nairobi. The reference survey map contains information on the registered property boundary details (beacons coordinates, dimension and area) of plot LR 7785/ 157, 158, 159 & 160 and the adjacent properties. The registered coordinates of Beacons **[H7, H8, H9, H43, H46 & H47**] are contained within the coordinate list in survey of Kenya reference map FR 131/69 top right corner with the mapping system for the coordinates being Cassini. The LOCHAB Property has a cumulative registered acreage of approximately 2.01 Hectares.

After obtaining the reference survey map, the survey team then moved to site to carry out the actual field exercise. While on site the team began by conducting initial boundary investigation which involved a physical search by digging up soil at the corners of the plot to expose any existing property beacons which could be underground. The team managed to locate Beacons H20 and H21 which were still in stable condition.

Using the 2 beacons identified above as a reference point, the survey team then used appropriate survey methods making use of the GNSS and Total Station equipment to re-establish the other boundary beacons **[H7, H8, H9, H43, H46 & H47**].

A typical boundary beacon appears in the form of an iron pin placed on the ground and secured by concrete (IPC). The approach was a success and the findings are documented in the FINDINGS section and captured in the drawing accompanying the report.

The Table below shows the beacon coordinates in both Cassini coordinate systems.

	Coordinate in Cassini System	
Point ID	Northings (M)	Eastings (M)
H7	-134400.79	-20471.12
H8	-134410.98	-20416.05
H9	-134417.26	-20360.40
H43	-134577.79	-20503.86
H46	-134589.07	-20442.89
H47	-134591.94	-20392.71

2. Detailed Survey

Thereafter, detail field survey commenced at the site. The team adopted Beacons H20 as benchmarks and referenced to UTM coordinate system to be used in the detailed survey. The features picked included and not limited to;

- Spot levels all over the open areas on site and inside some of adjacent properties
- Trees on and adjacent to site
- Permanent Structures at the site and on adjacent properties
- Temporary structures at the site
- Perimeter wall and Barbed wire fence
- Access road adjacent to the site e.g. Runda drive and Northern Bypass
- Dominant vegetation on site e.g. Trees
- Utility services on and off the site i.e. Overhead power line, Sewer lines on site
- Drainage infrastructure on and off the site

After fieldwork was complete, data was downloaded into field laptops for post processing. Processed data was then uploaded into drafting software i.e. AutoCAD. Final Topographic survey maps were then produced showing contours developed at intervals of 0.5m as per the Terms of Reference

Findings

In summary, we had the following findings and are explained as follows: -

1. The Plot Boundary

a) NONE of the 6 boundary beacons **[H7, H8, H9, H43, H46 & H47**] were found existing on site and as such all of them were re-established.

b) At Beacon **H7,** the existing perimeter wall corner is approx. 0.86m away from the beacon.

c) At Beacon **H8**, the existing perimeter wall corner is approx. 0.47m away from the beacon.

d) At Beacon **H9**, the existing perimeter wall corner is approx. 0.78m away from the beacon.

e) At Beacon **H43,** the existing perimeter wall corner is approx. 1.51m away from the beacon.

f) At Beacon **H46,** the existing perimeter wall corner is approx. 0.69m away from the beacon.

g) At Beacon **H47,** the beacon is to be located beneath an existing wall column.

h) In general, going by the position of the plot boundary vis-a-vis the existing perimeter wall, it then appears a section of the property land currently lies within

the neighboring properties which can either be as a result of incorrect construction of the existing perimeter wall and/or encroachment by the neighbours.

2. **The Perimeter wall**
   There is an existing perimeter wall constructed all around the property. The perimeter wall is approximately 3m high. This existing perimeter wall is not properly aligned to the property registered boundary.

3. **Utility Services**
   The only utility services available at the site are Power Supply and Water Supply. There exists a High tension overhead power line crossing into the property along boundary line H43 – H46. The other overhead power line is the one meant to supply power into the property. The water connection point is near the main gate as the survey captured water meter. Sewer facilities are onsite which consist of the use of septic tanks and soak pits.

4. **Access road**
   The property can only be accessed from Runda Drive which is at the frontage of the property. The road has a registered road reserve of 15m.

A topographic survey drawing accompanying this report will highlight all these findings and other features identified inside or adjacent to site.

Conclusion

As witnessed during the boundary survey exercise, the registered boundary of the plot appears to be crossing into existing properties. This could either be as a result of incorrect construction of the existing perimeter wall and/or encroachment by the neighbours.

My recommendation therefore is that, at some point before construction of the wall commences, the affected neighbours ought to be involved so that a joint survey is carried out so as to verify the cause of the encroachments if any.

TITLE

Topographic Survey of Properties LR 7785/157-160 of a cumulative acreage of 5 Acres along Runda Drive, Nairobi County

LEGEND

- Permanent Structures
- Temporary Structures
- Road edge (Tarmac)
- Walkway
- Storm Water drainage
- Culvert
- H20 Property Cadastral Boundary
- Gate
- Stone Wall
- Barbed Wire Fence
- Security lights
- Overhead Power Line
- Tree Canopy
- Flower
- Palm Tree
- Bamboo Plant
- CL (Cover Level) IL (Invert Level) Sewer Manhole
- CL (Cover Level) IL (Invert Level) Drainage Chamber
- 1705.22 Spot Level
- 1750.00 Contour line (V.I. = 0.5m)
- BM(Pin) 1705.14 Benchmark Position

N

Notes

1. All measurements are in meters.
2. Vertical datum is Mean sea Level
3. Coordinates referenced to UTM Arc 1960 Datum Zone 37 S
4. Reference Map is Survey of Kenya Map 131/69

CLIENT:

US EMBASSY
Nairobi, Kenya

SURVEYORS:

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Scale:	1:250
Date:	May 2021
Drawn by:	W.O.O